

The Art of Queueing

Reinforcement Learning for visitor flow at the Gallerie degli Uffizi in Florence

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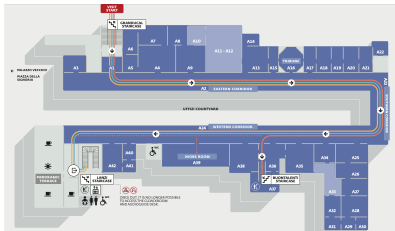
June 2026

The few rooms everyone queues for

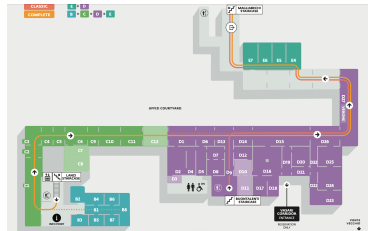
Almost all $\sim 5,000$ daily visitors come for the same handful of paintings, crammed into three of ninety-eight rooms, so those rooms run far over capacity while the rest sit calm. The crowd is mixed: 60% Instagram tourists, 30% normal, 10% art lovers.



Second floor (the gated masterpieces)



First floor (Caravaggio, free-flow)



The problem and the goal

Those three rooms run far over capacity while ninety-five sit half empty: the masterpiece rooms mean long queues and a crush; the rest, deserted galleries. The crowd never coordinates, so every visitor pressing into a packed Botticelli makes it worse for the next, a textbook *congestion externality*.

Our question: can reinforcement learning do better? Two goals.

The museum.

Redesign the incentives so the same crowd spreads out, raising welfare without turning anyone away or cutting revenue.

The visitor.

Find the smartest way through the museum despite the crowd: when to book, which masterpieces, how to pace the day.

Two streams, one project

We split the project into two parts kept separate until the end.

Stream 1: crowd simulator.

Full 60/30/10 crowd, one day.

- Museum *as-is* versus a Pigovian redesign.
- Adopt only if welfare rises, revenue holds.
- It passes: welfare **+31%**, revenue preserved.

Stream 2: RL on top (our focus).

Take that museum as fixed; find the *optimal* visitor.

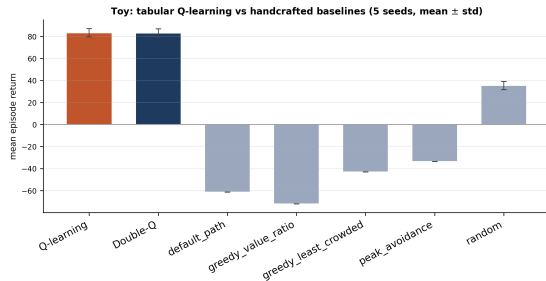
- When to book and which masterpieces.
- How to pace the day.
- All under crowd uncertainty.

Foundations: a tabular proof of concept

We prove it on a *12-room toy* with an exact value table. Crowds wave up and down, unannounced.

Tabular *Q-learning* learns *temporal arbitrage*: enter in the lull around the crowd peak.

Across five seeds: Q and *Double-Q* tie (83.3 vs 83.1). A deterministic toy has no overestimation bias to remove, exactly as theory predicts.



Learning beats every hand-written rule; the rules even go negative by mistiming the crowd.

Scaling up: deep RL and action masking

The real museum breaks the table: ninety-eight rooms, ~ 100 moves per step, hundreds of state numbers. We swap it for a neural net (*MaskablePPO*; PPO and DQN as controls).

Action masking: the rulebook

From any room only a few moves are legal, yet the net has ~ 100 outputs. We set every illegal move to $-\infty$ before the softmax, so it gets exactly zero probability. No gradient is wasted on moves that do not exist; this state-dependent action set is our single biggest design choice.

$$\pi(a \mid s) = \frac{e^{z_a} \mathbf{1}[a \in \mathcal{A}(s)]}{\sum_b e^{z_b} \mathbf{1}[b \in \mathcal{A}(s)]}.$$

As-is versus intervened: why it becomes a booking problem

We now leave the toy and put the agent on the *full museum*, running the same algorithms in two worlds.

- *As-is (the baseline)*: no reservations. You walk in and take the masterpieces as you find them, so with a map the route is a solved shortest path.
- *Intervened*: our *eleven Pigovian measures*. The headline one, *RAMA*, ends walk-in access: every masterpiece needs a slot *booked days ahead* (lead time 1, 7, 21 or 35).

That single rule flips the visitor's decision from *where to walk* into *when to book*, hard for three reasons walking never had: *uncertain* (how fast the good slots fill is hidden), *irreversible* (book late, the masterpiece is gone), *delayed* (you act today; the payoff lands at check-in, $\gamma = 0.999$).

The eleven interventions: pricing the externality

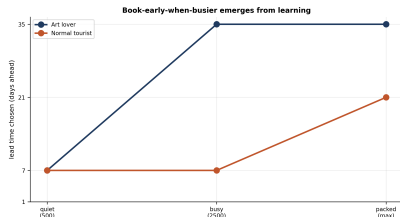
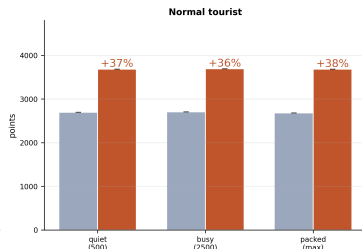
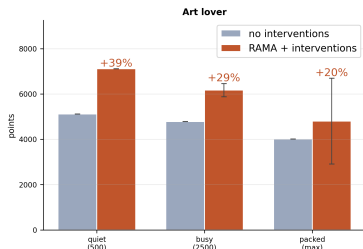
RAMA is one of *eleven* Pigovian measures behind the intervened museum. Slide three called the crush a congestion externality; the remedy is to make each visitor feel the cost they impose and reward those who spread out. For example:

- *Time*: extended opening hours (7 to 19), quiet-hour discounts, timed entry.
- *Access*: RAMA reservations, secondary-attractor enrichment to pull crowds off the masterpieces.
- *Price*: dynamic peak pricing, a flat surcharge on large tour groups.

A *director's gate* decides adoption: keep the bundle only if welfare rises and revenue holds. It does, **+31%** welfare with revenue preserved. Stream two then finds the optimal visitor inside this redesigned world.

Result: book early when busier and it pays

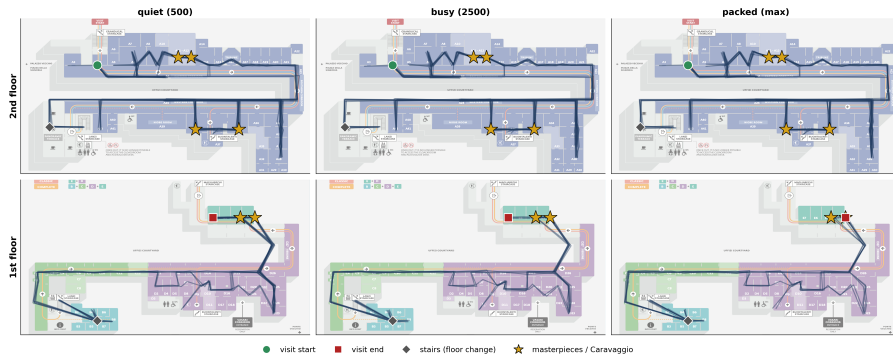
RAMA booking: intervened vs matched baseline (5 seeds, mean $\pm$ std; 3/3 masterpieces secured)



Under RAMA, both profiles beat the matched *as-is* (no-booking) baseline at every crowd level: **+20% to +39%** for the art lover, **+36% to +38%** for the tourist, all three masterpieces secured. The learned lead time *rises with the crowd* (right): book-early-when-busier is not coded in, it emerges from the reward, because booking late at a packed museum misses the rooms.

What a visit looks like, on the real floor plan

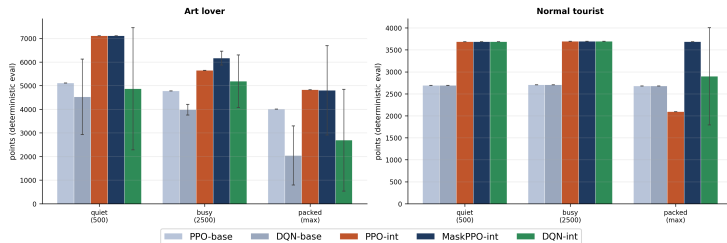
Art lover walking routes (RAMA + interventions) -- one continuous visit, 8 rollouts/panel; It descends floor 2 -> floor 1.



The art lover's learned visit, eight rollouts per panel, across the three crowd levels. It is one continuous, sensible route: enter at A1, sweep the second-floor masterpieces (gold stars) in their booked windows, descend the staircase to Caravaggio on the first floor and leave before closing. Every step is a one-hop move to a neighbouring room, so there are no teleports.

Which algorithm wins and why masking matters

Algorithm comparison: value-based (DQN) vs policy-gradient (PPO / MaskablePPO), baseline vs intervened (5 seeds, mean $\pm$ std)



- *MaskablePPO* is best or tied in every cell and stable across seeds.
- *Masking is decisive exactly at the hardest cell*: at the packed tourist, unmasked PPO *collapses* to 2093, below the no-booking baseline of 2680, while MaskablePPO holds 3685. At easier crowds the two agree; the gap appears only where precision matters.
- *DQN is crowd-fragile and high-variance* (its baseline falls 4527 $\rightarrow$ 2044): taking a max over noisy crowd values overestimates, the very bias the deterministic toy did not have.

1: an unlearnable reward

S *Symptom.* The agent never left, squatting in one room until closing. Zeroing all reward for not exiting made it *worse*.

D *Diagnosis.* An all-or-nothing wipe is *flat* almost everywhere, so the gradient is zero: nothing points to the door.

F *Fix.* Use a *dense* egress signal: per-step pressure rising near closing and far from an exit, plus a bonus for leaving.

L *Lesson.* Sparse rewards are correct but often unlearnable. Shape a slope the optimiser can climb.

2: it could not see what it was deciding

S *Symptom.* The rule “stay until satisfied, then move” never stabilised; dwell times wobbled at random.

D *Diagnosis.* The right action depended on value already taken from the room, *not in the state*: a hidden variable, a POMDP not the MDP we assumed.

F *Fix.* Add a per-room *appreciation-progress* value to the state, so “seen enough?” becomes something the policy can read.

L *Lesson.* The state must hold everything the optimal action depends on, or you solve a different problem.

3: reward shaping that backfired

S *Symptom.* Well-meant terms backfired: a “boredom” penalty caused *looping*, a harsh crowd penalty made a *fake art lover*.

D *Diagnosis.* Every term is a new objective to game (reward hacking): looping beat exiting, a crowded Botticelli scored worse than skipping it.

F *Fix.* *Minimal* shaping: a *gentle* crowd discount so a crowded masterpiece still beats skipping, no boredom penalty, satiation alone stops lingering.

L *Lesson.* Ask of every term: what is the cheapest way to collect it? Is that what you want?

4: the do-nothing trap and the book-early trick

S *Symptom*. Given a free “decline” button, *both* profiles declined all three masterpieces; book-early stayed only *half* learned.

D *Diagnosis*. Declining is safe now, booking pays only later: a do-nothing optimum; its reward lands hundreds of steps away (credit assignment).

F *Fix*. Make “always books” a *type trait* with no escape hatch; add an *immediate off-pace penalty* proxying the distant cost.

L *Lesson*. A small immediate signal tracking a far-off reward makes the long-horizon lesson learnable: our most decisive trick.

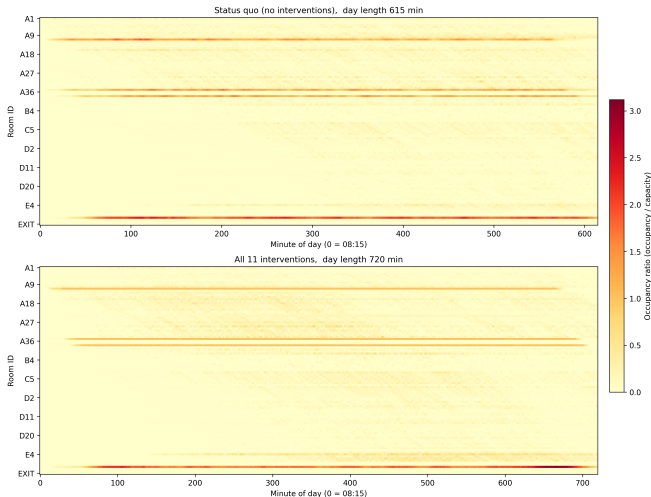
What five seeds revealed (and why we ran them)

A single training run can be lucky or unlucky, so a single number can mislead. We retrained every cell under *five* training seeds and report the mean and the spread across them.

- *The tourist is rock-stable*: essentially zero variance at every crowd level.
- *The art lover degrades as the museum fills*: the gain falls from +39% to +29% to +20%.
- *The catch*: at *maximum crowd* the art lover is *bimodal*, four of the five seeds reach $\sim +41\%$ but one **collapses** below baseline, so the cell averages +20% with a wide error bar. We report the spread, not the lucky seed.

The crowd before and after the redesign

Crowd density heatmap: status quo vs all 11 interventions (~5000 visitors). Shared colour scale.



Top: the museum as-is.

Bottom: all eleven interventions, same colour scale.

The masterpiece bands (A9, A36) cool from red toward orange and the day spreads over more hours.

+31% welfare, revenue held.

Thank you.

Questions?

Appendix

Backup slides for questions

Backup: the three MDPs at a glance

Setting	State	Action	Reward	γ
Toy (Q, Double-Q)	5-tuple table	stay / move	toy per-step	0.99
As-is (PPO, DQN)	8 numbers	stay / move on	walk, no booking	0.995/0.997
RAMA (MaskablePPO, PPO, DQN)	25 numbers	lead / book / pace masked for MPPO	booking per-step	0.999

The same five elements in every setting; what changes is the problem the agent is attached to, plus how the state is stored and the actions screened. PPO and DQN appear twice, once as a no-booking walk (the baselines) and once as booking agents; MaskablePPO only on RAMA.

Backup: the as-is state, eight numbers

The visitor's local view on the planned walk (natural units; the network sees each scaled to about $[0, 1]$):

- 1 *Room-value drain*: minutes absorbed vs the room's worth (~ 3 min per importance point).
- 2 *Room value*: importance, 1 to 10.
- 3 *Density*: how full the room is, percent of capacity.
- 4 *On-target flag*: are you in the next planned must-see?
- 5 *Plan completed*: share of must-sees finished.
- 6 *Time of day*: minute of the 615-minute day.
- 7 *Egress slack*: spare minutes to reach an exit before closing.
- 8 *Distance to next room*: doorways to the next must-see.

At the door A1: (0, 0.22, 0.30, 0, 0, 0, 0.99, 0.10).

Backup: the RAMA state, twenty-five numbers

A two-phase episode (a booking screen, then the walk), grouped:

- *Booking position* (3): phase flag, which of 4 decisions, lead chosen (days).
- *Booking screen* (4): slots free at lead 1/7/21/35 days (out of 8).
- *Time of day* (1): minute of the 720-minute day.
- *Windows held* (3): booked clock-times for Botticelli, Leonardo, Raphael.
- *Check-ins* (3): which masterpieces collected.
- *Access read* (4): is-masterpiece, unbooked, open-now, minutes-to-open.
- *Pace prior* (3): when it expects to reach each masterpiece (frozen).
- *Room tail* (4): drain, value, density, egress, as in the as-is state.

Opening screen: (1, 0, 1; 0.12, 0.38, 0.62, 1; 0; ...); the booking screen is the crux.

Actions per setting

- Toy / as-is: Discrete(2), stay or move on.
- RAMA: Discrete(4). Booking phase: a lead $\in \{1, 7, 21, 35\}$ or book/decline.
Walk phase: stay or move on.

Masking (MaskablePPO). Set the logits of illegal moves to $-\infty$ before the softmax, so they get exactly zero probability:

$$\pi(a \mid s) = \frac{e^{z_a} \mathbf{1}[a \in \mathcal{A}(s)]}{\sum_b e^{z_b} \mathbf{1}[b \in \mathcal{A}(s)]}.$$

Backup: the reward functions

Toy, per step. $\frac{\text{importance} \times \text{novelty}}{1 + \alpha_A \text{density}^2} - \text{step cost}$, with $\alpha_A = 6$, step 0.5, exit bonus +15, no-exit penalty -50 (hand-tuned shaping; the relative sizes are what matter).

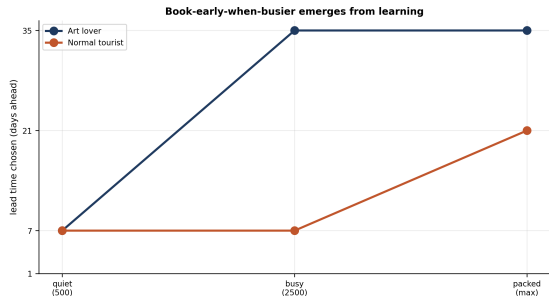
Full museum, per minute. $\text{importance} \times \text{crowd discount} \times \text{novelty (satiation)}$, with a gentle crowd $\alpha = 0.5$, plus closing-time pressure, a completion bonus and a small step cost.

Booking terms (RAMA only). an *immediate off-pace penalty* at commit time (a local proxy for the distant cost of a badly-timed slot) plus a no-show penalty for a masterpiece booked but missed.

Two-phase episode

- Booking (no time passes): pick a *lead time* (1/7/21/35 days), then book each masterpiece.
- Windows committed near the expected arrival; then the minute-by-minute walk.

Fill curve (busy day). Slots free out of 8: 1 a day before, 3 a week ahead, 5 three weeks ahead, 8 a month ahead. Book late, miss the masterpiece.



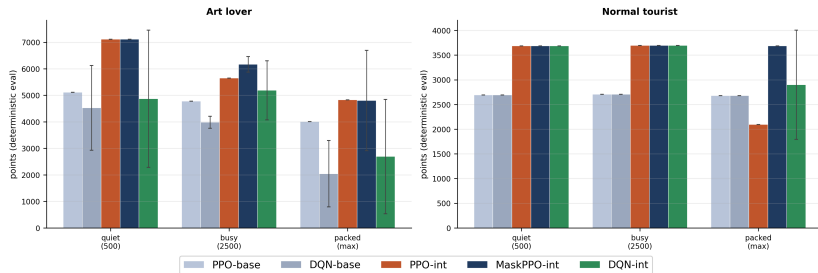
Backup: full booking results, five seeds (mean $\pm$ std)

Profile	Crowd	Baseline	Intervened	Gain	Check-ins
Art	500	5112	7113 ± 0	+39%	3.0/3
Art	2500	4780	6167 ± 291	+29%	3.0/3
Art	5000	4010	4800 ± 1891	+20%	2.6/3
Tourist	500	2694	3686 ± 0	+37%	3.0/3
Tourist	2500	2708	3695 ± 0	+36%	3.0/3
Tourist	5000	2680	3685 ± 0	+38%	3.0/3

Baseline = best no-booking walk (ppo_base); intervened = MaskablePPO on RAMA. Five training seeds {1, 3, 7, 42, 202606}, each evaluated on six fixed common-random-number crowds. The art lover at max crowd is bimodal: four seeds reach $\sim +41\%$, one collapses, hence the wide band and the 2.6/3 check-ins.

Backup: which algorithm and where it breaks

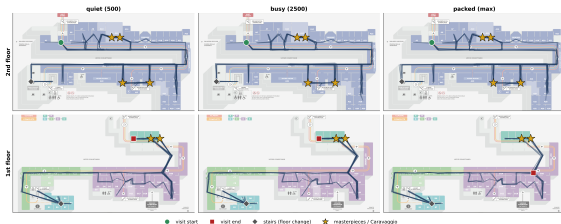
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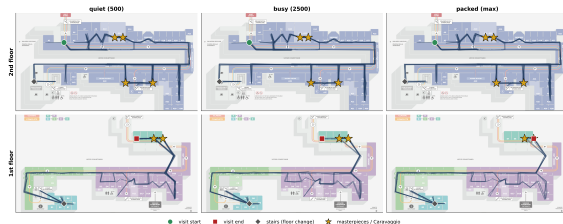
- *MaskablePPO*: best or tied in every cell, near-zero seed variance.
- *Unmasked PPO*: collapses at the packed tourist to 2093, below the 2680 no-booking baseline, while MaskablePPO holds 3685.
- *DQN*: crowd-fragile and high-variance, its art baseline falls 4527 $\rightarrow$ 2044 as the crowd grows (a max over noisy values overestimates).

Backup: the learned walk, art lover

Art lover walking routes (no interventions) – one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



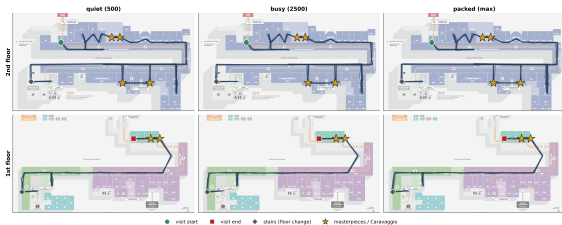
Art lover walking routes (RAMA + interventions) – one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



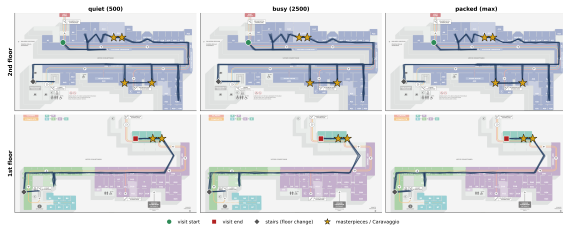
Left: as-is. Right: under RAMA. Eight rollouts per panel across the three crowd levels; the masterpieces (gold stars) are taken in their booked windows, then a descent to Caravaggio and out.

Backup: the learned walk, tourist

Normal tourist walking routes (no interventions) -- one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.

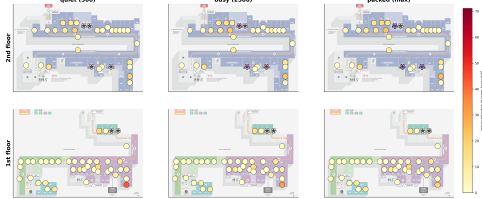


Normal tourist walking routes (RAMA + interventions) -- one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.

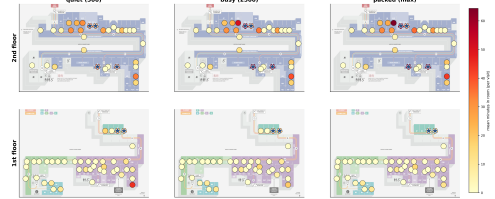


Backup: per-profile crowd density

Art lover -- where time is spent (no interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.



Art lover -- where time is spent (RAMA + interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.



Normal tourist -- where time is spent (no interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.

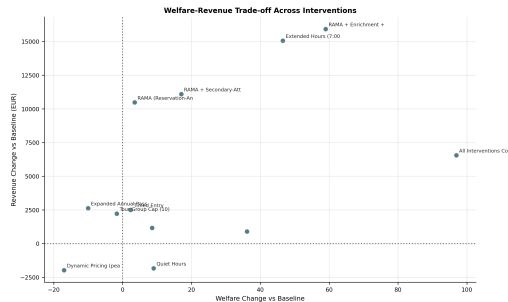


Normal tourist -- where time is spent (RAMA + interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.

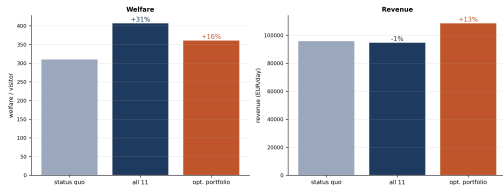


Left column as-is, right column under RAMA; art lover on top, tourist below. The booked windows spread each profile's masterpiece visits across the day.

Backup: the interventions and the director's gate

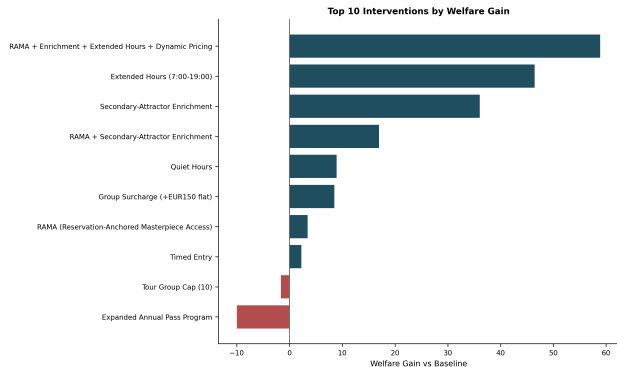


Museum-wide director gate (as-is crowd): all 11 lift welfare with revenue preserved; the optimized subset raises both



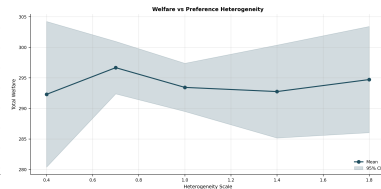
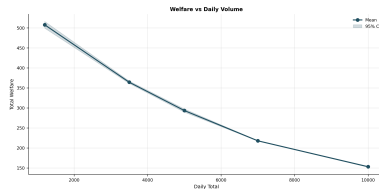
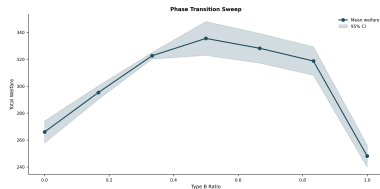
The redesign is adopted only if welfare rises and revenue does not fall (the director's gate). The chosen portfolio of eleven Pigovian measures lifts welfare **+31%** with revenue preserved.

Backup: ranking the candidate interventions by welfare gain



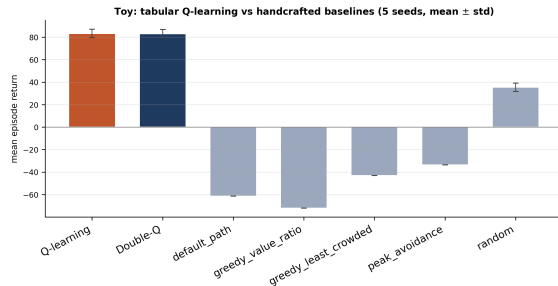
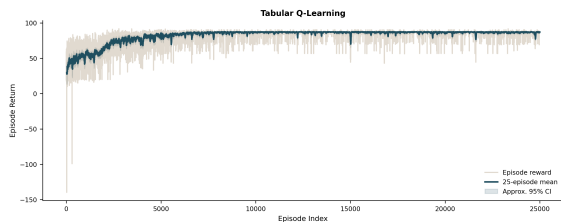
Each candidate measure scored on its own against the as-is baseline. Extended opening hours and secondary-attractor enrichment are the strongest single levers; the full combined bundle leads. The tour-group cap and the expanded annual pass lower welfare on their own; the adopted portfolio keeps the welfare-positive measures.

Backup: robustness sweeps



We vary the Type-B (Instagram) share, the total daily volume and the crowd heterogeneity. The welfare gain from the redesign holds across the ranges we tried.

Backup: tabular toy internals



Left: the learning curve, return climbing over episodes. Right: learning beats every hand-written rule. Across five seeds Q (83.3 ± 3.8) and Double-Q (83.1 ± 3.7) tie, since a deterministic toy has no overestimation bias for Double-Q to remove.

Deep RL (Stable-Baselines3). MlpPolicy [128, 128], Adam LR 3×10^{-4} , 2048 steps, batch 256, 10 epochs, GAE $\lambda = 0.98$, clip 0.2, entropy 0.01.

- As-is baselines (PPO): 150k steps, γ 0.995/0.997.
- RAMA arms: 100k steps, $\gamma = 0.999$.

Toy. tabular Q / Double-Q, ε -greedy, $\gamma = 0.99$, 25k episodes.

Seeds and evaluation

- 5 training seeds {1, 3, 7, 42, 202606}.
- Eval on 6 fixed crowds (900000 to 900005), deterministic, common random numbers.
- Crowd levels 500 / 2500 / 5000 visitors a day.

Backup: limitations and where we would go next

One visitor, not a crowd. What we optimise is a single sophisticated visitor's best response to a *fixed* crowd. It is not an equilibrium: if every visitor booked this way the good slots would fill faster and the fill curve itself would move. Closing that loop, a mean-field game over the whole population, is the real next step.

Fixes we did not run. Our toy isolates the overestimation that makes DQN crowd-fragile, yet we never tried the obvious cure, Double-DQN, on the full task. Two experiments would sharpen the story: a fixed-heuristic booker (“always reserve a month out”) as a tougher baseline than no-booking, plus an ablation that switches the off-pace penalty off to confirm that booking early is learned rather than coaxed.

What is baked in. The connoisseur always books (a trait, not a trick to win); lead times are four discrete values; the fill curve is calibrated rather than fitted to real booking data; the policy is a plain feedforward net carrying appreciation-progress, with $\gamma = 0.999$. Welfare, finally, is the simulator's own visitor-utility, itself a modelling choice rather than a settled social objective.